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The synthesis of polysubstituted indoles from 3-bromo-2-indolyl phosphates†

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A novel methodology for the synthesis of functionalised indoles based on the cross-coupling reactions of 3-bromo-2-indolyl phosphates is described. The preparation involves the conversion of easily available 2-oxindoles to 3,3-dibromo-2-oxindoles followed by the Perkow reaction affording 3-bromo-2-indolyl phosphates. Then bromine atom is substituted regioselectively by the Suzuki coupling reaction. We observed that aluminum chloride promoted the reaction of 3-substituted-2-indolyl phosphates with organozinc reagents furnishing 2,3-disubstituted indoles as final products. The overall diversity and efficiency of the methodology was demonstrated by the synthesis of bioactive molecule from easily available substances.

Introduction

The substitution of a carbon atom by a heteroatom in aromatic compounds significantly influences the properties and reactivity of the heterocyclic compounds compared to their parent carbon-only analogues. Indole is a representative example of such a structure. Besides other practical indole applications, many naturally occurring compounds contain the indole skeleton,¹ and indole derivatives are associated with significant bioactivity.² The importance of substances containing the indole motif has resulted in the elaboration of numerous synthetic protocols to access substituted indoles. Methodologies suitable for the preparation of substituted indoles can be divided into two groups. The first type utilises cyclisation reactions of substituted benzenes.³ The other synthetic approach is based on the modification of easily available indole building blocks. The key reaction frequently used is C–H activation.⁴ A complementary approach relies on the traditional cross-coupling reactions of pure electrophilic templates represented by halogenated indoles. Although dichloroindoles can be prepared by chlorination of indole with S₂Cl₂,⁵ *N*-chloro-*N*-methoxybenzenesulfonamide,⁶ their transition metal catalysed reactions are rare. On the other hand, the vast majority of cross-coupling processes are associated with brominated indoles. Dibromo-, tribromo-, and tetrabromo indoles are usually synthesised by the reaction of the indole with bromine,⁷ NBS,⁸ or 1,3-dibromo-5,5-dimethylhydanthoin.⁹ Marginal reactivity studies involved the Heck reaction,⁸ regioselective debromina-

tion,^{7a} and the Suzuki reaction¹⁰ of tetrabromo-, and tribromoindoles. In the series of 2,3-dibromoindoles, the Heck reaction,⁸ debromination,¹¹ and the Negishi¹² reaction have been reported. Significant effort has been devoted to the course of the Suzuki reaction of 2,3-dibromoindoles.^{10,13}

In sharp contrast to aryl and vinyl halides, the corresponding substituted phosphates represent a very attractive starting substrate¹⁴ for divergence-oriented synthesis for several reasons. Firstly, aryl and vinyl phosphates are easily preparable. Secondly, the low reactivity of phosphates in cross-coupling reactions compared to halides indicated such phosphates for highly regioselective introductions of various substituents. Although the cross-coupling reactions of phosphates heading to substituted indoles were reported,¹⁵ there is no systematic study on the preparation and reactivity of halogenated 2-indolyl phosphates. The only report describes the preparation of *p*-toluidine 5-bromo-6-chloro-3-indolyl phosphate as a substrate for cytochemical demonstration of enzyme activity.¹⁶ Thus, based on our previous results obtained during the chemo- and stereoselective cross-coupling of dibromoenoil phosphates,¹⁷ we decided to devise a methodology for the synthesis of 3-bromo-2-indolyl phosphate, and explore its reactivity.

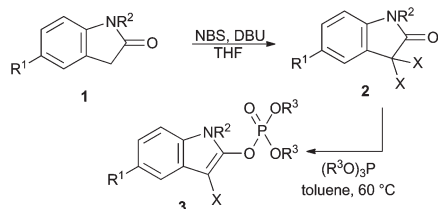
Results and discussion

At this point, we envisioned that a strategy based on the conversion of 2-oxindoles to 3,3-dibromo-2-oxindoles followed by the Perkow reaction would fit our goal to synthesise phosphates **3** (Table 1). 3,3-Dibromo-2-oxindoles can be prepared by reaction of 2-oxindoles with CuBr₂,¹⁸ Br₂ in CCl₄,¹⁹ radical bromination,²⁰ NBS in acetonitrile²¹ and pyridine·HBr·Br₂.²²

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Table 1 Preparation of 3-halo-2-indolyl phosphates 3



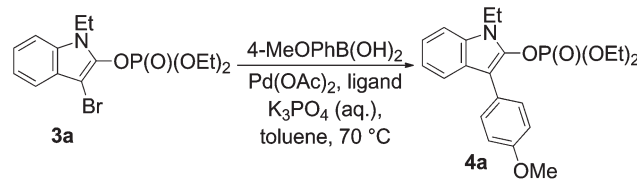
Entry	X	R ¹	R ²	R ³	2 ^a (%)	3 ^a (%)
1	Br	H	Et	Et	2a, 95	3a, 99
2	Cl	H	Et	Et	2b, 96	3b, 91
3	Br	H	Et	H ₂ C(CH ₃) ₂	—	3c, 82
4	Br	H	Boc	Ph	2d, 87	3d, 93
5	Br	F	Boc	Ph	2e, 72	3e, 92
6	Br	Cl	Boc	Ph	2f, 70	3f, 91
7	Br	Br	Boc	Ph	2g, 75	3g, 84
8	Br	MeO	Boc	Ph	2h, 90	3h, 87
9	Br	F	H	Ph	2i, 83	3i, 84
10	Br	Br	H	Ph	2j, 43	3j, 78

^a Isolated yield.

NBS-mediated bromination of 2-oxindoles was the method of choice for the preparation of 3,3-dibromo-2-oxindoles but the reported procedure²¹ suffers by the fact that initial bromination proceeds in position 5 followed by bromination in the position 3, limiting the universality of the protocol. However, we envisioned that the presence of base would favour the bromination in position 3. Indeed, our bromination of 1-ethyl-2-oxindole afforded 3,3-dibromo indole 2a as the sole product in very high yield (Table 1, entry 1). The methodology also enables the preparation of chlorinated product 2b in 96% yield (Table 1, entry 2). Boc-protected bromoindoles 2d–h were also prepared in high yields (Table 1, entries 4–8). Moreover, we were able to prepare unprotected 3,3-dibromooxindoles 2i,j in good yields (Table 1, entries 9 and 10). In the next step we focused on the preparation of phosphates 3 via the Perkow reaction. Prepared 3,3-dibromo-2-oxindoles were treated with triethyl or diphenyl(ethyl) phosphite in dry toluene at 60 °C. In contrast to the low reactivity of amides in the Perkow reaction,²³ the reaction of 3,3-dihalo-2-oxindoles with phosphites turned out to be high-yielding, affording diethyl, and diphenyl phosphates 3a,b,d–h (Table 1, entries 1, 2 and 4–8). Cyclic phosphate 3c, and unprotected phosphates 3i,j were synthesized in good isolated yield (Table 1, entries 3, 9 and 10). Thus, the two step procedure gave the phosphates 3a–i in average overall yields ranged from 63 to 94% with the exception of phosphate 3j that was isolated in 34% overall isolated yield.

With phosphates 3a–h in our hands we started to study their behaviour in cross-coupling reactions. The reaction of diethyl-1-ethyl-3-bromo-2-indolyl phosphate (3a) with 4-methoxyphenylboronic acid catalysed with palladium was a reaction of choice. We realised that TPP, tricyclohexylphosphine, and

Table 2 Optimisation of the Suzuki reaction of phosphate 3a

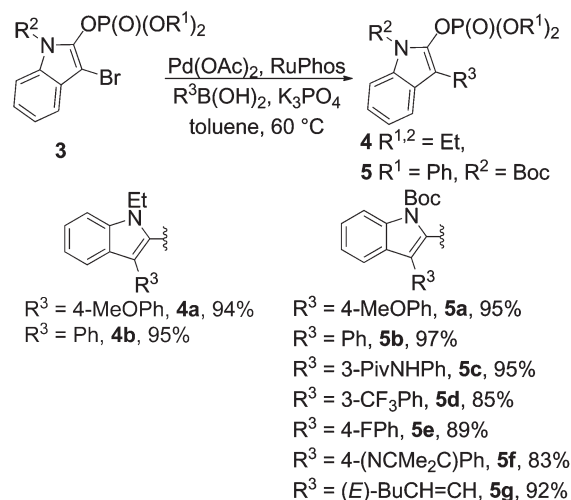


Entry	Time [h]	Ligand	Cat./ligand loading [mol%]	Conv. (%)
1	16	PPh ₃	5/10	15
2	16	Cy ₃ P	5/10	61
3	16	JohnPhos	5/10	50
4	28	SPhos	5/10	90 (85 ^a)
5	7	RuPhos	5/10	100 (94 ^a)
6	14	RuPhos	2/4	100 (94 ^a)

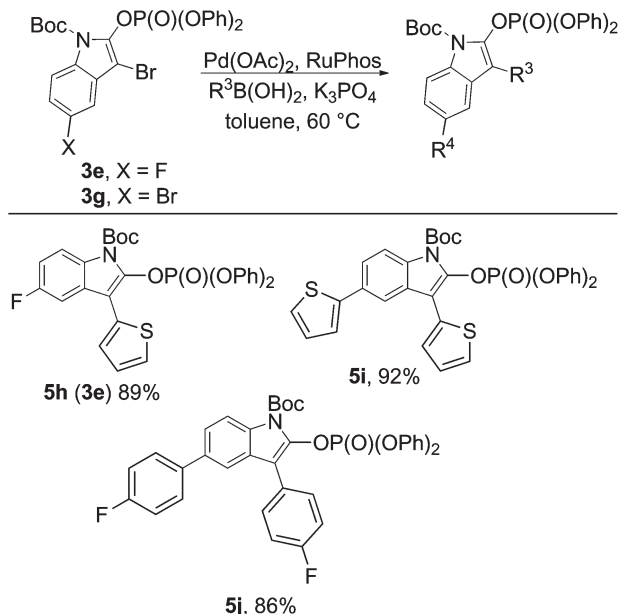
^a Isolated yield.

JohnPhos contributed poorly to the reaction course giving no more than 61% conversion of 3a (Table 2, entries 1–3). SPhos increased the conversion to 90% within 28 h reaction time (Table 2, entry 4). On the other hand, complete conversion of 3a to 4a was achieved with 5 mol% of Pd(OAc)₂, and 10 mol% of RuPhos (Table 2, entry 5). Decreasing catalyst loading to 2 mol% prolonged the reaction time to 14 h maintaining 94% isolated yield (Table 2, entry 6).

Optimised reaction conditions were used to evaluate the scope of the developed protocol. Simple phenylboronic acids or boronic acids bearing amide, trifluoromethyl, fluoro and nitrile functionalities coupled with starting bromoindoles affording 3-substituted indoles 4b, 5a–f in almost quantitative yields (Scheme 1). The methodology also enables stereo-selective introduction of alkenyl moiety as demonstrated on the preparation of indole 5g in 92% yield. The coupling reaction of unprotected phosphates 3i,j were unsuccessful leading



Scheme 1 Structure of prepared 3-substituted 2-indolyl phosphates 4 and 5.



Scheme 2 The cross-coupling reaction of 3,5-dihaloindoles **3e,g**.

to complex reaction mixture. In case of 2-indolyl phosphates **3e–g** bearing halogen in the 5 position, significant selectivity was achieved for substrates bearing a fluorine atom **5h** (Scheme 2). The other phosphates, with chlorine and bromine atoms, coupled unselectively, although they offered access to 3,5-disubstituted indoles **5i,j** (Scheme 2).

The phosphate **5a** was used as a model compound to study the substitution of phosphate moiety. The Negishi reaction of **5a** was a method of choice based on our positive experiences with aluminum chloride-promoted coupling of alkenyl phosphates with organozinc reagents under mild conditions.^{17b} Thus, **5a** was treated with 4-methylphenylzinc chloride in the presence of $\text{PdCl}_2(\text{SPhos})_2$ in dry THF at 45 °C (Table 3). When

1.5 eq. of 4-methylphenylzinc chloride was coupled with **5a** without AlCl_3 , negligible conversion of starting compound was observed along with significant amount of decomposition of **5a** (Table 3, entry 1). Analysis of the crude reaction mixture by ^1H NMR revealed that phosphate conversion to 2-oxindole, and Boc deprotection are the main pathways responsible for the above-mentioned decomposition. Increased equivalents of organozinc reagent and addition of aluminum chloride led to 75% isolated yield of **6a** with 90% conversion (Table 3, entries 2–4). $\text{PdCl}_2\text{XantPhos}$ did not improve the conversion of **5a**; however PdCl_2DPPE gave 100% conversion of **5a** in 91% isolated yield (Table 3, entries 6 and 7). The Boc group turned out to be sufficiently stable under tested conditions (Table 3, entry 7), however the reaction time prolongation resulted in partial Boc group deprotection. It is worth mentioning that experiments with diethyl phosphates **4a** were not successful, and decomposition of **4a** to 2-oxindole and other unidentified by-products, was observed under the above-tested conditions.

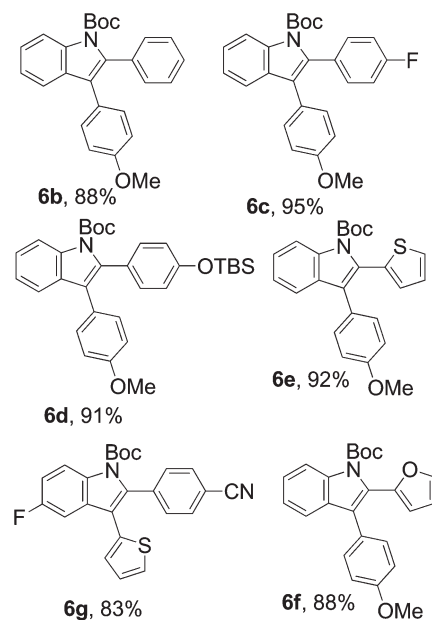
Representative examples of prepared 2,3-disubstituted indoles are shown in Scheme 3. As can be seen, the developed method enables introduction of both simple aryl substituents, and 2-thienyl fragment **6b–e**, in excellent yields. The use of aluminum chloride did not interfere with Lewis acid-sensitive 2-furyl substituent **6f**, which was obtained in 88% isolated yield. Functionalised organozinc reagent bearing a nitrile group **6g** also coupled smoothly with starting phosphate **5h**.

To demonstrate synthetic power of the developed methodology we decided to regioselectively prepare the indole derivative **7** (Scheme 4), which was reported as a compound with significant leishmanicidal activity.²⁴ The synthesis of compound **7** started from 3-bromo-2-indolyl phosphate **3h**, which coupled with commercially available 3,4,5-trimethoxyphenylboronic

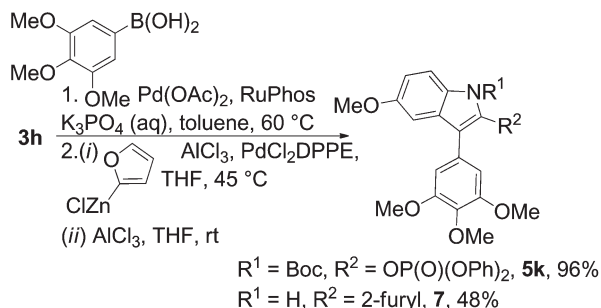
Table 3 Optimisation of coupling of phosphate **5a** with 4-methylphenylzinc chloride mediated by aluminum chloride

Entry	Pd-cat.	RZnCl [eq.]	AlCl_3 [eq.]	Time [h]	Conv. [%]
1	$\text{PdCl}_2(\text{SPhos})_2$	1.5	0	3	<5 ^a
2	$\text{PdCl}_2(\text{SPhos})_2$	1.5	0.75	5	58
3	$\text{PdCl}_2(\text{SPhos})_2$	2.5	0.75	5	81
4	$\text{PdCl}_2(\text{SPhos})_2$	3.0	1.0	24	90(75 ^b)
5	$\text{PdCl}_2(\text{SPhos})_2$	1.5	1.5	3	90(64 ^b)
6	$\text{PdCl}_2\text{XantPhos}$	3.0	1.0	24	60
7	PdCl_2DPPE	3.0	1.0	5	100(91 ^b)

^a Decomposition of **5a** was observed. ^b Isolated yield.



Scheme 3 Structure of prepared 2,3-disubstituted indoles by the aluminum chloride-mediated Negishi reaction.



Scheme 4 Accomplished synthesis of the bioactive molecule **7**.

acid giving phosphate **5k** in 96% isolated yield as a single regioisomer. To minimise the number of operations during the preparation of the product **7** we took the advantage of the low stability of Boc-protected indoles in the presence of aluminium chloride. Thus, phosphate **5k** was coupled with 2-furylzinc chloride in the presence of 1.0 eq. of aluminium chloride, the solution was concentrated and addition of 3.0 eq. of AlCl_3 in THF removed the Boc protecting group affording product **7** in 48% isolated yield.

Conclusions

In conclusion, we have developed a novel methodology for regioselective synthesis of 2,3-disubstituted indoles. The developed procedure consists of several key steps. Initially 3,3-dibromo-2-oxindoles were prepared by improved regioselective bromination of 2-oxindole with NBS in the presence of DBU. Isolated brominated products were heated with trialkyl phosphites in dry toluene affording products of the Perkow reaction, which is rare in the case of α -bromoamides. Then we utilised the different reactivities of activated C–O and C–Br bonds in cross-coupling reactions, and the first substituents were introduced to position 3 by the Suzuki coupling reaction. Next, aluminium chloride-promoted cross-coupling reaction of 3-substituted-2-indolyl phosphates with organozinc reagents completed the synthesis of 2,3-disubstituted indoles. The scope of the developed methodology was demonstrated on a limited number of examples; however, we believe that a combination of easily available substituted 2-oxindoles and the overall robustness of the protocol has potential to supply a wide variety of bioactive compounds.

Experimental

General information

All reactions were performed under an argon atmosphere. NMR spectra were measured on a Varian Gemini 300 (^1H , 300.07 MHz; ^{13}C , 75.46 MHz), spectrometer at 298 K. Mass spectra were measured on ZAB-SEQ (VG Analytical). The dry and degassed solvents were prepared by PureSolv MD7; silica gel (Merck, Silica Gel 60, 40–63 μm) was used for column

chromatography. BuLi (2.5 M solution in hexane), and other compounds were purchased. Concentration of BuLi was determined by titration using menthol and 1,10-phenanthroline before use. Organozinc reagents were prepared by the standard procedure starting from bromo- or iodoarenes (1.0 eq.), butyllithium (1.1 eq.) in dry THF ($-80\text{ }^\circ\text{C}$, 30 min), followed by the transmetalation with ZnCl_2 (1.0 eq., $0\text{ }^\circ\text{C}$, 30 min).

General procedure for the preparation of dihalogen-2-oxindoles **2**

NBS (20.5 mmol) or NCS (20.5 mmol) was added in one portion to a solution of 2-oxindole **1** (10 mmol) in anhydrous THF (40 mL) cooled to $0\text{ }^\circ\text{C}$. Then DBU (0.22 mL, 1.5 mmol) was added dropwise *via* syringe, and the deep red solution formed was stirred for 1 h at $0\text{ }^\circ\text{C}$. The cooling bath was removed, and the reaction was stirred for another 40 min. The reaction mixture was diluted with ether (50 mL), the organic phase washed with water (40 mL) and brine (40 mL), dried over MgSO_4 , filtered and concentrated *in vacuo*. Column chromatography (silicagel) afforded the final products.

3,3-Dibromo-1-ethyl-2-oxindole (2a). Yield 95% (3.04 g); hexane : EtOAc (6 : 1, R_f 0.4); m.p.: $95\text{--}96\text{ }^\circ\text{C}$; ^1H NMR (300 MHz, CDCl_3): δ 7.62 (d, $J = 7.7\text{ Hz}$, 1H), 7.33 (t, $J = 7.8\text{ Hz}$, 1H), 7.16 (t, $J = 7.5\text{ Hz}$, 1H), 6.82 (d, $J = 8.1\text{ Hz}$, 1H), 3.78 (q, $J = 7.2\text{ Hz}$, 3H), 1.30 (t, $J = 7.2\text{ Hz}$, 3H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 169.4, 138.9, 131.5, 131.2, 126.1, 123.9, 109.1, 45.4, 35.8, 12.2; HRMS (EI) $[M]^+$ calcd for $\text{C}_{10}\text{H}_9\text{Br}_2\text{NO}$: 316.9052, found: 316.9046.

3,3-Dichloro-1-ethyl-2-oxindole (2b). Yield 96% (2.21 g); hexane : EtOAc (6 : 1, R_f 0.5); ^1H NMR (300 MHz, CDCl_3): δ 7.62 (d, $J = 8.1\text{ Hz}$, 1H), 7.39 (t, $J = 7.8\text{ Hz}$, 1H), 7.17 (t, $J = 7.7\text{ Hz}$, 1H), 6.87 (d, $J = 7.7\text{ Hz}$, 1H), 3.79 (q, $J = 7.3\text{ Hz}$, 2H), 1.30 (t, $J = 7.4\text{ Hz}$, 3H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 168.5, 139.8, 131.8, 129.4, 125.0, 123.9, 109.2, 36.5, 35.7, 12.3; HRMS (EI) $[M]^+$ calcd for $\text{C}_{10}\text{H}_9\text{Cl}_2\text{NO}$: 229.0061, found: 229.0058.

3,3-Dibromo-1-(tert-butyloxycarbonyl)-2-oxindole (2d). Yield 87% (3.41 g); hexane : EtOAc (20 : 1, R_f 0.6); m.p.: $107\text{--}109\text{ }^\circ\text{C}$; ^1H NMR (300 MHz, CDCl_3): δ 7.84 (d, $J = 8.1\text{ Hz}$, 1H), 7.67 (d, $J = 7.5\text{ Hz}$, 1H), 7.41–7.38 (m, 1H), 7.30–7.25 (m, 1H), 1.66 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 167.1, 148.4, 136.0, 131.9, 130.0, 126.0, 125.8, 115.6, 85.8, 45.1, 28.0; HRMS (ESI) $[M + \text{Na}]^+$ calcd for $\text{C}_{13}\text{H}_{13}\text{Br}_2\text{NO}_3$: 413.9134, found: 413.9133.

3,3-Dibromo-1-(tert-butyloxycarbonyl)-5-fluoro-2-oxindole (2e). Yield 72% (2.95 g); hexane $\rightarrow$ hexane : DCM (9 : 1) $\rightarrow$ hexane : DCM (4 : 1, R_f 0.3); m.p.: $125\text{--}128\text{ }^\circ\text{C}$; ^1H NMR (300 MHz, CDCl_3): δ 7.87 (dd, $J = 4.2, 9.0\text{ Hz}$, 1H), 7.38 (dd, $J = 3.0, 6.9\text{ Hz}$, 1H), 7.09 (dt, $J = 3.0, 9.0\text{ Hz}$, 1H), 1.65 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 166.7, 160.2 (d, $J = 246.5\text{ Hz}$), 148.4, 132.0 (d, $J = 2.6\text{ Hz}$), 131.4 (d, $J = 8.6\text{ Hz}$), 118.8 (d, $J = 22.7\text{ Hz}$), 117.3 (d, $J = 8.0\text{ Hz}$), 113.1 (d, $J = 25.6\text{ Hz}$), 86.0, 44.1, 28.0; ^{19}F NMR (CDCl_3): δ -115.4 (m); HRMS (ESI) $[M + \text{Na}]^+$ calcd for $\text{C}_{13}\text{H}_{12}\text{Br}_2\text{FNO}_3$: 431.9040, found: 431.9042.

3,3-Dibromo-1-(tert-butyloxycarbonyl)-5-chloro-2-oxindole (2f). Yield 70% (2.99 g); hexane $\rightarrow$ hexane : EtOAc (20 : 1, R_f 0.55); m.p. $125\text{--}130\text{ }^\circ\text{C}$; ^1H NMR (300 MHz, CDCl_3): δ 7.83

(d, J = 8.7 Hz, 1H), 7.64 (d, J = 1.8 Hz, 1H), 7.35 (dd, J = 2.4, 8.7 Hz, 1H), 1.65 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 166.4, 148.2, 134.5, 131.9, 131.4, 131.2, 126.0, 117.0, 86.1, 43.7, 28.0; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{13}\text{H}_{12}\text{Br}_2\text{ClNO}_3$: 447.8744, found: 447.8748.

3,3,5-Tribromo-1-(tert-butyloxycarbonyl)-2-oxindole (2g). Yield 75% (3.52 g); hexane : DCM (9 : 1) $\rightarrow$ hexane : EtOAc (9 : 1, R_f 0.65); m.p.: 110–112 °C; ^1H NMR (300 MHz, CDCl_3): δ 7.79–7.76 (m, 2H), 7.52–7.48 (m, 1H), 1.65 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 166.3, 148.2, 135.0, 134.8, 131.7, 128.9, 118.5, 117.3, 86.2, 43.5, 28.0; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{13}\text{H}_{12}\text{Br}_3\text{NO}_3$: 491.8239, found: 491.8240.

3,3-Dibromo-1-(tert-butyloxycarbonyl)-5-methoxy-2-oxindole (2h). Yield 90% (3.78 g); hexane : EtOAc (20 : 1, R_f 0.4); m.p.: 94–95 °C; ^1H NMR (300 MHz, CDCl_3): δ 7.73 (d, J = 9.6 Hz, 1H), 7.15 (d, J = 3.0 Hz, 1H), 6.89 (dd, J = 8.7, 3.0 Hz, 1H), 3.83 (s, 3H), 1.63 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 167.0, 157.5, 148.4, 130.7, 129.0, 117.9, 116.7, 110.6, 85.4, 55.7, 45.5, 27.9; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{14}\text{H}_{15}\text{Br}_2\text{NO}_4$: 443.9240, found: 443.9242.

3,3-Dibromo-5-fluoro-2-oxindole (2i). Yield 83% (2.56 g); hexane : EtOAc (3 : 1, R_f 0.55); m.p.: >128 °C (decomp); ^1H NMR (300 MHz, CDCl_3): δ 9.67 (br s, 1H), 7.35–7.31 (m, 1H), 7.04–6.96 (m, 2H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 172.4, 159.6 (d, J = 244.2 Hz), 133.0 (d, J = 2.5 Hz), 132.7 (d, J = 8.7 Hz), 118.6 (d, J = 23.7 Hz), 113.7 (d, J = 26.0 Hz), 109.8 (d, J = 7.7 Hz), 44.6; ^{19}F NMR (CDCl_3): δ -117.6 (m); HRMS (ESI) $[\text{M}]^-$ calcd for $\text{C}_8\text{H}_3\text{Br}_2\text{FNO}$: 307.8550, found: 307.8552.

3,3,5-Tribromo-2-oxindole (2j). Acetonitrile (40 mL) was used as the solvent; yield 43% (1.59 g); hexane : EtOAc (3 : 1, R_f 0.3); m.p.: >124 °C (decomp); ^1H NMR (300 MHz, CDCl_3): δ 8.59 (br s, 1H), 7.73 (s, 1H), 7.43 (dd, J = 1.8, 8.7 Hz, 1H), 6.83 (d, J = 8.7 Hz, 1H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 171.2, 135.8, 134.5, 133.2, 129.2, 116.7, 112.6, 43.7; HRMS (ESI) $[\text{M}]^-$ calcd for $\text{C}_8\text{H}_3\text{Br}_3\text{NO}$: 367.7750, found: 367.7749.

General procedure for preparation of (3-bromo-2-indolyl) phosphates 3

Triethyl phosphite, ethyl(diphenyl) phosphite, or 2-methoxy-5,5-dimethyl-1,3,2-dioxaphosphorinane (8.4 mmol) was dropwise added to a solution of 3,3-dihalogen-2-oxindole 2 (7 mmol) in dry toluene (28 mL). The reaction mixture was stirred at 60 °C for designated time period. Then the solvent was evaporated under reduce pressure and column chromatography afforded the target phosphates 3. Some phosphates were isolated by crystallization: the reaction mixture was evaporated *in vacuo* to about 1/3 of its original volume and diluted with hexane to the original volume. The precipitated phosphates were isolated by filtration. Next portion of crystals was obtained by the filtrate concentration and dilution with hexane.

(3-Bromo-1-ethyl-2-indolyl)diethyl phosphate (3a). 12 h; yield 99% (2.60 g); hexane : EtOAc (6 : 1) $\rightarrow$ hexane : EtOAc (3 : 1, R_f 0.25); ^1H NMR (300 MHz, CDCl_3): δ 7.52–7.49 (m, 1H), 7.24–7.18 (m, 3H), 4.36–4.23 (m, 6H), 1.42–1.36 (m, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 138.4 (d, J = 8.0 Hz), 130.4,

125.0 (d, J = 1.7 Hz), 122.0 (d, J = 1.0 Hz), 120.3, 118.6, 109.2, 75.4 (d, J = 6.1 Hz), 65.4 (d, J = 6.1 Hz), 37.6, 15.8 (d, J = 6.7 Hz), 14.4; HRMS (ESI) $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{14}\text{H}_{19}\text{BrNO}_4\text{P}$: 376.0308, found: 376.0314.

(3-Chloro-1-ethyl-2-indolyl)diethyl phosphate (3b). 12 h; yield 91% (2.11 g); hexane : EtOAc (6 : 1) $\rightarrow$ hexane : EtOAc (3 : 1, R_f 0.25); m.p.: 35–37 °C; ^1H NMR (300 MHz, CDCl_3): δ 7.56 (d, J = 7.5 Hz, 1H), 7.27–7.18 (m, 3H), 4.38–4.20 (m overlap with q, 6H), 1.42–1.36 (m, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 136.9 (d, J = 8.3 Hz), 129.7, 123.8, 122.2, 120.4, 117.9, 109.4, 89.9 (d, J = 5.0 Hz), 65.6 (d, J = 6.0 Hz), 37.6, 16.0 (d, J = 6.6 Hz), 14.7; HRMS (ESI) $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{14}\text{H}_{19}\text{ClNO}_4\text{P}$: 332.0813, found: 332.0818.

(3-Bromo-1-ethyl-2-indolyl)neopentylglycol phosphate (3c). 12 h; yield 82% (2.20 g); m.p.: 136–141 °C; ^1H NMR (300 MHz, CDCl_3): δ 7.49 (d, J = 7.8 Hz, 1H), 7.26–7.18 (m, 3H), 4.60 (d, J = 10.4 Hz, 2H), 4.27 (q, J = 7.5 Hz, 2H), 4.13 (d, J = 11.1 Hz, 1H), 4.06 (d, J = 11.1 Hz), 1.41 (t, J = 7.5 Hz, 3H), 1.37 (s, 3H), 0.98 (s, 3H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 138.6 (d, J = 8.0 Hz), 130.8, 125.1 (d, J = 1.7 Hz), 122.3, 120.6, 118.8, 109.5, 79.0 (d, J = 7.4 Hz), 75.3 (d, J = 5.8 Hz), 38.0, 32.2 (d, J = 6.2 Hz), 21.7, 20.0 (d, J = 1.1 Hz), 14.7; HRMS (ESI) $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{15}\text{H}_{19}\text{BrNO}_4\text{P}$: 388.0308, found: 388.0317.

(3-Bromo-1-tert-butyloxycarbonyl-2-indolyl)diphenyl phosphate (3d). 1 h; yield 93% (3.55 g); m.p.: 113–115 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.07 (d, J = 8.1 Hz, 1H), 7.51–7.48 (m, 1H), 7.39–7.28 (m, 6H), 7.27–7.15 (m, 6H), 1.64 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 150.4 (d, J = 7.0 Hz), 148.1, 137.5 (d, J = 9.8 Hz), 131.4 (d, J = 1.2 Hz), 129.8, 125.8 (d, J = 2.3 Hz), 125.7 (d, J = 2.3 Hz), 125.4 (d, J = 1.1 Hz), 123.6, 120.1 (d, J = 5.0 Hz), 119.2, 115.2 (d, J = 1.1 Hz), 86.2 (d, J = 8.1 Hz), 85.5, 28.0; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{25}\text{H}_{23}\text{BrNO}_6\text{P}$: 566.0339, found: 566.0339.

(3-Bromo-1-tert-butyloxycarbonyl-5-fluoro-2-indolyl)diphenyl phosphate (3e). 1 h; yield 92% (3.61 g); m.p.: 126–128 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.05 (dd, J = 3.9, 9.3 Hz, 1H), 7.36–7.31 (m, 4H), 7.26–7.15 (m, 7H), 7.07 (dt, J = 9.2, 2.6 Hz, 1H), 1.63 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 159.7 (d, J = 241.1 Hz), 150.3 (d, J = 7.5 Hz), 147.8, 138.5 (d, J = 9.5 Hz), 129.8, 127.6, 127.0 (d, J = 2.4 Hz), 126.8 (d, J = 2.3 Hz), 125.7 (d, J = 1.7 Hz), 120.0 (d, J = 5.1 Hz), 116.8 (d, J = 8.8 Hz), 113.2 (d, J = 24.8 Hz), 105.2 (d, J = 25.6 Hz), 85.9, 28.0; ^{19}F NMR (CDCl_3): δ -118.9 (m); HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{25}\text{H}_{22}\text{BrFNO}_6\text{P}$: 584.0244, found: 584.0246.

(3-Bromo-1-tert-butyloxycarbonyl-5-chloro-2-indolyl)diphenyl phosphate (3f). 1 h; yield 91% (3.68 g); m.p.: 108–111 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.01 (d, J = 8.7 Hz, 1H), 7.48 (d, J = 2.4 Hz, 1H), 7.36–7.29 (m, 5H), 7.26–7.20 (m, 6H), 1.63 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 150.3 (d, J = 7.4 Hz), 147.7, 138.4 (d, J = 9.5 Hz), 129.8, 129.7 (d, J = 1.1 Hz), 129.5 (d, J = 1.0 Hz), 127.0 (d, J = 2.3 Hz), 125.8 (d, J = 1.7 Hz), 125.6 (d, J = 1.1 Hz), 120.0 (d, J = 4.9 Hz), 118.9, 116.6, 86.0, 85.3 (d, J = 7.5 Hz), 28.0; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{25}\text{H}_{22}\text{BrClNO}_6\text{P}$: 599.9949, found: 599.9950.

(3,5-Dibromo-1-tert-butyloxycarbonyl-2-indolyl)diphenyl phosphate (3g). 1 h; yield 84% (3.66 g); m.p.: 123–124 °C;

^1H NMR (300 MHz, CDCl_3): δ 7.96 (d, J = 9.3 Hz, 1H), 7.63 (d, J = 1.5 Hz, 1H), 7.45 (dd, J = 1.8, 9.3 Hz, 1H), 7.36–7.31 (m, 4H), 7.26–7.18 (m, 6H), 1.63 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 150.3 (d, J = 7.6 Hz), 147.7, 138.2 (d, J = 9.8 Hz), 130.1 (d, J = 1.1 Hz), 129.8, 128.3, 127.4 (d, J = 2.5 Hz), 125.8 (d, J = 1.6 Hz), 122.0 (d, J = 1.0 Hz), 120.0 (d, J = 5.1 Hz), 117.0 (d, J = 1.1 Hz), 116.9 (d, J = 1.0 Hz), 86.1, 85.1 (d, J = 7.4 Hz), 28.0; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{25}\text{H}_{22}\text{Br}_2\text{NO}_6\text{P}$: 645.9423, found: 645.9426.

(3-Bromo-1-*tert*-butyloxycarbonyl-5-methoxy-2-indolyl)diphenyl phosphate (3h). 1 h; yield 87% (3.50 g); m.p.: 106–109 °C; ^1H NMR (300 MHz, CDCl_3): δ 7.97 (d, J = 8.6 Hz, 1H), 7.37–7.28 (m, 4H), 7.27–7.15 (m, 6H), 6.98–6.92 (m, 2H), 3.88 (s, 3H), 1.63 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ = 156.6, 150.4 (d, J = 7.0 Hz), 148.0, 137.8 (d, J = 9.8 Hz), 129.8, 126.6 (d, J = 3.2 Hz), 125.8, 125.6, 120.1 (d, J = 5.2 Hz), 116.4, 114.3, 101.7, 86.1 (d, J = 1.2 Hz), 85.3, 55.7, 28.0; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{26}\text{H}_{25}\text{BrNO}_7\text{P}$: 596.0444, found: 596.0447.

(3-Bromo-5-fluoro-2-indolyl)diphenyl phosphate (3i). 1 h; yield 84% (2.71 g); m.p.: 137–139 °C; ^1H NMR (300 MHz, CDCl_3): δ 9.20 (br s, 1H), 7.40–7.35 (m, 4H), 7.28–7.22 (m, 6H), 7.13–7.08 (m, 2H), 6.94–6.88 (m, 1H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 158.6 (d, J = 237.0 Hz), 149.9 (d, J = 8.0 Hz), 139.4 (d, J = 7.7 Hz), 130.0 (d, J = 1.1 Hz), 129.9, 126.7, 126.3 (d, J = 1.7 Hz), 120.1 (d, J = 4.6 Hz), 112.6 (d, J = 9.4 Hz), 111.1 (d, J = 26.1 Hz), 104.0 (d, J = 25.4 Hz), 75.4 (dd, J = 9.2, 4.5 Hz); ^{19}F NMR (CDCl_3): δ –122.3 (m); HRMS (ESI) $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{20}\text{H}_{14}\text{BrNFO}_4\text{P}$: 463.9880, found: 463.9883.

(3,5-Dibromo-2-indolyl)diphenyl phosphate (3j). 1 h; yield 78% (2.83 g); hexane:EtOAc (9:1) $\rightarrow$ hexane:EtOAc (3:1, R_f 0.55); m.p.: 140–142 °C; ^1H NMR (300 MHz, CDCl_3): δ 9.73 (br s, 1H), 7.58 (d, J = 1.8 Hz, 1H), 7.38–7.19 (m, 11H), 6.95 (d, J = 9.0 Hz, 1H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 149.8 (d, J = 8.0 Hz), 139.0 (d, J = 7.7 Hz), 130.1 (d, J = 1.1 Hz), 128.9, 127.2, 126.3 (d, J = 1.5 Hz), 125.7, 121.2, 120.1 (d, J = 4.6 Hz), 114.3, 113.1, 74.8 (d, J = 8.7 Hz); HRMS (ESI) $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{20}\text{H}_{14}\text{Br}_2\text{NO}_4\text{P}$: 523.9080, found: 523.9071.

General procedure for the Suzuki cross-coupling reaction of phosphates 3

Toluene (4 mL) was added to a mixture of 3-bromoindol-2-yl phosphate 3 (1 mmol), 2 mol% $\text{Pd}(\text{OAc})_2$ (0.02 mmol), 4 mol% RuPhos (0.04 mmol) and boronic acid or ester (1.4 mmol). The mixture was stirred for 5 minutes at the ambient temperature. Then 1 M solution of K_3PO_4 in water (3 mL, 3 eq.) was added and the reaction mixture was stirred for indicated time period, and temperature. Then the mixture was diluted with Et_2O (20 mL), washed with saturated NH_4Cl (10 mL) solution, water (10 mL) and brine (10 mL), dried over MgSO_4 , and column chromatography (silicagel) afforded the final products.

[1-Ethyl-3-(4-methoxyphenyl)-2-indolyl]diethyl phosphate (4a). 14 h; 70 °C; yield 94% (0.380 g); hexane:EtOAc (3:1, R_f 0.3); ^1H NMR (300 MHz, CDCl_3): δ 7.70 (d, J = 8.1 Hz, 1H), 7.57 (dd, J = 2.1, 6.9 Hz, 2H), 7.34–7.14 (m, 3H), 7.00 (dd, J = 2.1, 6.9 Hz, 2H), 4.30 (q, J = 7.2 Hz, 2H), 4.04–3.99 (m, 2H), 3.88–3.82 (m, 2H), 1.46 (t, J = 7.2 Hz, 3H), 1.19–1.14 (m, 6H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 158.0, 137.7 (d, J = 8.9), 131.1,

130.2, 125.7, 125.6, 121.5, 120.1, 119.2, 113.9, 109.3, 101.1, 65.0 (d, J = 6.6 Hz), 55.3, 37.3, 15.9 (d, J = 7.2 Hz), 14.7; HRMS (ESI) $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{21}\text{H}_{26}\text{NO}_5\text{P}$: 404.1621, found: 404.1622.

(1-Ethyl-3-phenyl-2-indolyl)diethyl phosphate (4b). 16 h; 70 °C; yield 95% (0.355 g); hexane:EtOAc (3:1, R_f 0.3); ^1H NMR (300 MHz, CDCl_3): δ 7.75 (d, J = 7.8 Hz, 1H), 7.67–7.64 (m, 2H), 7.48–7.43 (m, 2H), 7.36–7.14 (m, 4H), 4.32 (q, J = 7.2 Hz, 2H), 4.06–4.00 (m, 2H), 3.87–3.79 (m, 2H), 1.48 (t, J = 7.2 Hz, 3H), 1.17–1.12 (m, 6H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 138.0 (d, J = 8.8 Hz), 133.3, 131.2, 129.2, 128.4, 126.0, 125.1, 121.6, 120.2, 119.2, 109.4, 101.5, 65.0 (d, J = 6.1 Hz), 37.3, 15.8 (d, J = 6.6 Hz), 14.7; HRMS (ESI) $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{20}\text{H}_{24}\text{NO}_4\text{P}$: 374.1516, found: 374.1518.

[1-*tert*-Butyloxycarbonyl-3-(4-methoxyphenyl)-2-indolyl]diphenyl phosphate (5a). 10 h; 60 °C; yield 95% (0.545 g); hexane:EtOAc (6:1, R_f 0.3); m.p.: 115–117 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.16 (d, J = 8.1 Hz, 1H), 7.60 (d, J = 7.8 Hz, 1H), 7.50–7.47 (m, 2H), 7.36–7.12 (m, 9H), 6.97–6.94 (m, 3H), 6.88–6.85 (m, 2H), 3.80 (s, 3H), 1.69 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 158.8, 150.3 (d, J = 7.5 Hz), 148.9, 135.7 (d, J = 11.0 Hz), 132.0 (d, J = 1.2 Hz), 130.6 (d, J = 1.2 Hz), 129.5, 126.4 (d, J = 2.5 Hz), 125.2 (d, J = 1.1 Hz), 124.5, 123.2, 123.1 (d, J = 2.2 Hz), 119.7 (d, J = 5.1 Hz), 119.4 (d, J = 1.2 Hz), 115.1, 114.0, 108.6 (d, J = 6.5 Hz), 84.8, 55.2, 28.1; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{32}\text{H}_{30}\text{NO}_7\text{P}$: 594.1652, found: 594.1651.

(1-*tert*-Butyloxycarbonyl-3-phenyl-2-indolyl)diphenyl phosphate (5b). 20 h; 60 °C; yield 97% (0.525 g); hexane:EtOAc (6:1, R_f 0.4); m.p.: 104–106 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.16 (d, J = 8.1 Hz, 1H), 7.63 (d, J = 7.5 Hz, 1H), 7.59–7.55 (m, 2H), 7.39–7.09 (m, 13H), 6.92–6.89 (m, 2H), 1.67 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 150.3 (d, J = 7.4 Hz), 148.8, 136.0 (d, J = 11.0 Hz), 132.1 (d, J = 1.3 Hz), 130.9 (d, J = 2.0 Hz), 129.6 (m, two overlapping signals), 128.5, 127.3, 126.2 (d, J = 2.8 Hz), 125.3 (d, J = 1.1 Hz), 124.6, 123.2, 119.8 (d, J = 5.0 Hz), 119.4 (d, J = 1.0 Hz), 115.1, 108.9 (d, J = 6.3 Hz), 85.0, 28.1; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{31}\text{H}_{28}\text{NO}_6\text{P}$: 564.1547, found: 564.1549.

[1-*tert*-Butyloxycarbonyl-3-(3-pivaloylamidophenyl)-2-indolyl]diphenyl phosphate (5c). 6 h; 60 °C; yield 84% (0.540 g); hexane:EtOAc (6:1, R_f 0.3); m.p.: 63–65 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.15 (d, J = 8.4 Hz, 1H), 7.69–7.61 (m, 2H), 7.49 (s, 1H), 7.38–7.08 (m, 12H), 6.95–6.92 (m, 3H), 1.68 (s, 9H), 1.29 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 176.3, 150.2 (d, J = 7.5 Hz), 148.7, 138.3, 136.0 (d, J = 11.0 Hz), 132.0 (d, J = 1.1 Hz), 131.4 (d, J = 1.9 Hz), 129.5, 129.1, 126.0 (d, J = 2.5 Hz), 125.4 (d, J = 1.0 Hz), 125.2, 124.6, 123.3, 120.8 (d, J = 1.2 Hz), 119.6 (d, J = 5.2 Hz), 119.4, 119.1, 115.0, 108.5 (d, J = 6.3 Hz), 85.0, 39.5, 28.0, 27.5; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{36}\text{H}_{37}\text{N}_2\text{O}_7\text{P}$: 663.2231, found: 663.2230.

[1-*tert*-Butyloxycarbonyl-3-(3-trifluoromethylphenyl)-2-indolyl]diphenyl phosphate (5d). 13 h; 60 °C; yield 85% (0.510 g); hexane:EtOAc (9:1, R_f 0.35); ^1H NMR (300 MHz, CDCl_3): δ 8.17 (d, J = 8.1 Hz, 1H), 7.85 (s, 1H), 7.72 (d, J = 7.5 Hz, 1H), 7.56 (d, J = 7.8 Hz, 1H), 7.51–7.36 (m, 3H), 7.31–7.09 (m, 7H), 6.92–6.89 (m, 4H), 1.69 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3):

δ 150.1 (d, J = 7.5 Hz), 148.6, 136.5 (d, J = 10.8 Hz), 132.8, 132.1 (d, J = 1.1 Hz), 131.9 (d, J = 2.0 Hz), 130.9 (q, J = 32.4 Hz), 129.6, 129.0, 126.1 (dq, J = 3.8, 1.7 Hz), 125.6 (d, J = 2.7 Hz), 125.4 (d, J = 1.5 Hz), 124.9 (d, J = 1.1 Hz), 124.0 (q, J = 27.2 Hz), 123.9 (q, J = 3.9 Hz), 123.5, 119.5 (d, J = 5.1 Hz), 118.9, 115.3, 107.4 (d, J = 6.3 Hz), 85.3, 28.1; HRMS (ESI) $[M + Na]^+$ calcd for $C_{32}H_{27}F_3NO_6P$: 632.1420, found: 632.1420.

[1-*tert*-Butyloxycarbonyl-3-(4-fluorophenyl)-2-indolyl]diphenyl phosphate (5e). 8 h; 60 °C; yield 89% (0.50 g); hexane : EtOAc (9 : 1, R_f 0.45); 1H NMR (300 MHz, $CDCl_3$): δ 8.16 (d, J = 8.7 Hz, 1H), 7.58–7.45 (m, 3H), 7.40–7.33 (m, 1H), 7.30–7.19 (m, 5H), 7.17–7.10 (m, 2H), 7.04–6.92 (m, 6H), 1.69 (s, 9H); ^{13}C NMR (75.5 MHz, $CDCl_3$): δ 162.0 (d, J = 246.7 Hz), 150.2, (d, J = 7.3 Hz), 148.8, 136.1 (d, J = 11.0 Hz), 132.0 (d, J = 1.7 Hz), 131.2 (dd, J = 8.0, 1.7 Hz), 129.6, 126.8 (dd, J = 3.4, 1.7 Hz), 126.1 (d, J = 2.8 Hz), 125.4 (d, J = 1.1 Hz), 124.7, 123.3, 119.7 (d, J = 5.5 Hz), 119.1 (d, J = 1.0 Hz), 115.5 (d, J = 21.4 Hz), 115.2, 107.9 (d, J = 6.4 Hz), 85.1, 28.1; ^{19}F NMR ($CDCl_3$): δ -115.1 (m); HRMS (ESI) $[M + Na]^+$ calcd for $C_{31}H_{27}FNO_6P$: 582.1452, found: 582.1450.

{1-*tert*-Butyloxycarbonyl-3-[4-(2-cyanoprop-2-yl)phenyl]-2-indolyl} diphenyl phosphate (5f). 12 h; 60 °C; yield 83% (0.505 g); hexane : EtOAc (9 : 1) $\rightarrow$ hexane : EtOAc (6 : 1, R_f 0.4); 1H NMR (300 MHz, $CDCl_3$): δ 8.16 (d, J = 8.1 Hz, 1H), 7.59–7.57 (m, 3H), 7.43–7.37 (m, 3H), 7.27–7.13 (m, 8H), 6.92–6.89 (m, 3H), 1.69 (s, 9H); ^{13}C NMR (75.5 MHz, $CDCl_3$): δ 150.2 (d, J = 7.5 Hz), 148.7, 140.2, 136.2 (d, J = 11.2 Hz), 132.0 (d, J = 1.2 Hz), 130.6 (d, J = 2.3 Hz), 130.0 (d, J = 1.2 Hz), 129.6, 125.9 (d, J = 2.3 Hz), 125.4 (d, J = 1.1 Hz), 125.2, 124.7, 124.4, 123.3, 119.6 (d, J = 5.4 Hz), 119.2, 115.2, 108.0 (d, J = 6.3 Hz), 85.1, 36.9, 29.1, 28.1; HRMS (ESI) $[M + Na]^+$ calcd for $C_{35}H_{33}N_2O_6P$: 631.1968, found: 631.1959.

(*E*)-[1-*tert*-Butyloxycarbonyl-3-(hex-1-enyl)-2-indolyl]diphenyl phosphate (5g). 11 h; 60 °C; yield 92% (0.50 g); hexane : EtOAc (9 : 1 R_f 0.5); 1H NMR (300 MHz, $CDCl_3$): δ 8.14 (d, J = 7.9, 1H), 7.80 (d, J = 7.6, 1H), 7.38–7.28 (m, 6H), 7.27–7.16 (m, 6H), 6.50 (dd, J = 16.3, 1.1, 1H), 6.34 (dt, J = 16.3, 6.7, 1H), 2.14 (q, J = 6.8, 2H), 1.66 (s, 9H), 1.45–1.24 (m, 4H), 0.88 (t, J = 7.1, 3H); ^{13}C NMR (75.5 MHz, $CDCl_3$): δ 150.4 (d, J = 7.4 Hz), 148.7, 136.1 (d, J = 11.1 Hz), 133.4 (d, J = 2.3 Hz), 132.3 (d, J = 1.1 Hz), 129.7, 125.5 (d, J = 1.2 Hz), 125.3 (d, J = 2.6 Hz), 124.3, 123.1, 120.0 (d, J = 4.9 Hz), 119.9, 118.3 (d, J = 3.3 Hz), 115.1, 106.0 (d, J = 6.5 Hz), 84.7, 33.5, 31.4, 28.0, 22.2, 13.9; HRMS (ESI) $[M + Na]^+$ calcd for $C_{31}H_{34}NO_6P$: 570.2016, found: 570.2014.

[1-*tert*-Butyloxycarbonyl-5-fluoro-3-(2-thienyl)-2-indolyl] diphenyl phosphate (5h). 16 h; 60 °C; yield 89% (0.505 g); hexane : EtOAc (9 : 1) $\rightarrow$ hexane : EtOAc (6 : 1, R_f 0.45); m.p.: 124–126 °C; 1H NMR (300 MHz, $CDCl_3$): δ 8.10 (dd, J = 4.5, 9.3 Hz, 1H), 7.47 (dd, J = 2.4, 8.7 Hz, 1H), 7.33–7.25 (m, 6H), 7.17–6.99 (m, 8 H), 1.68 (s, 9H); ^{13}C NMR (75.5 MHz, $CDCl_3$): δ 159.7 (d, J = 239.8 Hz), 150.2 (d, J = 7.6 Hz), 148.4, 137.1 (d, J = 11.0 Hz), 131.1 (d, J = 2.2 Hz), 129.9 (d, J = 1.1 Hz), 129.7, 128.1 (dd, J = 1.1, 1.1 Hz), 127.3, 127.1 (d, J = 1.7 Hz), 125.9 (d, J = 1.5 Hz), 125.48 (d, J = 1.1 Hz), 125.45, 119.8 (d, J = 5.2 Hz), 116.4 (d, J = 8.8 Hz), 112.5 (d, J = 24.8 Hz), 105.5 (d, J = 24.9 Hz),

85.5, 28.0; ^{19}F NMR ($CDCl_3$): δ -119.5 (m); HRMS (ESI) $[M + Na]^+$ calcd for $C_{29}H_{25}FNO_6PS$: 588.10156, found: 588.10164.

[1-*tert*-Butyloxycarbonyl-3,5-bis(2-thienyl)-2-indolyl]diphenyl phosphate (5i). 17 h; 60 °C; yield 92% (0.581 g); hexane : EtOAc (9 : 1) $\rightarrow$ hexane : EtOAc (6 : 1, R_f 0.3); 1H NMR (300 MHz, $CDCl_3$): δ 8.13 (d, J = 8.7 Hz, 1H), 8.02 (d, J = 1.2 Hz, 1H), 7.62 (dd, J = 2.4, 8.7 Hz, 1H), 7.34–7.20 (m, 8H), 7.17–6.97 (m, 8H), 1.69 (s, 9H); ^{13}C NMR (75.5 MHz, $CDCl_3$): δ 150.3 (d, J = 7.6 Hz), 148.5, 144.7, 136.7 (d, J = 10.7 Hz), 131.34, 131.30 (d, J = 1.1 Hz), 130.1, 129.7, 128.0, 127.3, 127.2 (d, J = 1.3 Hz), 126.2 (d, J = 2.9 Hz), 125.5 (m, two overlapping signals), 124.5, 123.3, 122.9, 119.8 (d, J = 5.2 Hz), 116.9 (d, J = 1.2 Hz), 115.5, 103.0 (d, J = 6.4 Hz), 85.4, 28.0; HRMS (ESI) $[M + Na]^+$ calcd for $C_{33}H_{28}NO_6PS_2$: 652.0988, found: 652.0986.

[1-*tert*-Butyloxycarbonyl-3,5-bis(4-fluorophenyl)-2-indolyl] diphenyl phosphate (5j). 15 h; 60 °C; yield 86% (0.562 g); hexane : EtOAc (9 : 1) $\rightarrow$ hexane : EtOAc (6 : 1, R_f 0.5); 1H NMR (300 MHz, $CDCl_3$): δ 8.22 (d, J = 8.1 Hz, 1H), 7.67 (s, 1H), 7.59–7.48 (m, 5H), 7.28–7.21 (m, 4H), 7.18–7.08 (m, 4H), 7.06–6.96 (m, 6H), 1.71 (s, 9H); ^{13}C NMR (75.5 MHz, $CDCl_3$): δ 162.3 (d, J = 246.3 Hz), 162.08 (d, J = 246.9 Hz), 150.2 (d, J = 7.5 Hz), 148.7, 137.5 (d, J = 3.0 Hz), 136.5 (d, J = 10.5 Hz), 135.8, 131.3 (d, J = 1.3 Hz), 131.2 (dd, J = 8.1, 1.2 Hz), 129.6, 128.8 (d, J = 8.1 Hz), 126.64 (d, J = 2.5 Hz), 126.60 (d, J = 2.8 Hz), 125.4 (d, J = 1.2 Hz), 124.0, 119.6 (d, J = 5.2 Hz), 117.4, 115.59 (d, J = 21.4 Hz), 115.56, 115.54 (d, J = 21.4 Hz), 108.1 (d, J = 6.3 Hz), 85.2, 28.1; ^{19}F NMR ($CDCl_3$): δ -114.8 (m), -116.7 (m); HRMS (ESI) $[M + Na]^+$ calcd for $C_{37}H_{30}F_2NO_6P$: 676.1671, found: 676.1664.

[1-*tert*-Butyloxycarbonyl-3-(3,4,5-trimethoxyphenyl)-5-methoxy-2-indolyl]diphenyl phosphate (5k). 20 h; 60 °C; yield 96% (0.635 g); hexane : EtOAc (9 : 1) $\rightarrow$ hexane : EtOAc (3 : 1, R_f 0.4); 1H NMR (300 MHz, $CDCl_3$): δ 8.05 (d, J = 9.1 Hz, 1H), 7.27–7.18 (m, 4H), 7.16–7.06 (m, 3H), 7.00–6.93 (m, 5H), 6.75 (s, 2H), 3.82 (s, 3H), 3.81 (s, 3H), 3.74 (s, 6H), 1.68 (s, 9H); ^{13}C NMR (75.5 MHz, $CDCl_3$): δ 156.2, 153.2, 150.2 (d, J = 7.4 Hz), 148.7, 137.2, 136.2 (d, J = 10.9 Hz), 129.5, 126.9 (d, J = 2.7 Hz), 126.4 (d, J = 1.6 Hz), 126.2 (d, J = 2.3 Hz), 125.3 (d, J = 1.2 Hz), 119.5 (d, J = 5.2 Hz), 116.2, 112.9, 108.8 (d, J = 6.3 Hz), 106.6 (d, J = 1.6 Hz), 102.3, 84.8, 60.7, 59.9, 55.6, 28.0; HRMS (ESI) $[M + Na]^+$ calcd for $C_{35}H_{36}NO_{10}P$: 684.1969, found: 684.1968.

General procedure for the Negishi coupling of phosphates 5

Freshly prepared and titrated arylzinc chloride-LiCl reagent (1.5 mmol) was quickly added to phosphate 5 (0.5 mmol) and $PdCl_2DPPE$ complex (5 mol%). Then the mixture was warmed to 45 °C followed by addition of a 0.5 M solution of $AlCl_3$ in THF (0.5 mmol). The reaction was stirred for indicated time period, the reaction mixture was diluted with ether (10 mL) and a 1 M solution of Rochelle salt in water (3 mL) was added. The mixture was stirred for 20 minutes then the organic phase was washed with brine and dried over $MgSO_4$. Column chromatography (silicagel) afforded the final products.

1-*tert*-Butyloxycarbonyl-3-(4-methoxyphenyl)-2-(4-methylphenyl)indole (6a). 4.5 h; yield 91% (0.188 g); hexane : EtOAc (9 : 1, R_f 0.7); m.p.: 113–115 °C; 1H NMR (300 MHz, $CDCl_3$):

δ 8.27 (d, J = 8.4 Hz, 1H), 7.56 (d, J = 6.9 Hz, 1H), 7.40–7.34 (m, 1H), 7.28–7.22 (m, 1H), 7.17–7.09 (m, 6H), 6.84–6.81 (m, 2H), 3.80 (s, 3H), 2.36 (s, 3H), 1.29 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 158.2, 150.3, 137.0, 135.5, 135.6, 131.1, 130.9, 130.1, 129.6, 128.4, 125.7, 124.4, 122.8, 121.3, 119.5, 115.1, 113.6, 83.1, 55.1, 27.5, 21.3; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{27}\text{H}_{27}\text{NO}_3$: 436.1883, found: 436.1886.

1-tert-Butyloxycarbonyl-3-(4-methoxyphenyl)-2-phenylindole (6b). 5.5 h; yield 88% (0.176 g); hexane : EtOAc (9 : 1, R_f 0.7); m.p.: 151–153 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.31 (d, J = 8.4 Hz, 1H), 7.58 (d, J = 8.1 Hz, 1H), 7.39–7.36 (m, 1H), 7.30–7.26 (m, 6H), 7.16 (d, J = 8.4 Hz, 2H), 6.83 (d, J = 8.4 Hz, 2H), 3.80 (s, 3H), 1.27 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 158.3, 150.3, 136.5, 135.4, 134.1, 131.1, 130.3, 129.5, 127.7, 127.3, 125.6, 124.6, 122.9, 121.5, 119.6, 115.2, 113.6, 83.2, 55.1, 27.5; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{26}\text{H}_{25}\text{NO}_3$: 422.1727, found: 422.1728.

1-tert-Butyloxycarbonyl-2-(4-fluorophenyl)-3-(4-methoxyphenyl)indole (6c). 4 h; yield 95% (0.198 g); hexane $\rightarrow$ hexane : EtOAc (9 : 1, R_f 0.65); m.p. 158–161 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.31 (d, J = 8.4 Hz, 1H), 7.58 (d, J = 8.1 Hz, 1H), 7.40 (t, J = 7.9 Hz, 1H), 7.31–7.23 (m, 3H), 7.14 (d, J = 8.6 Hz, 2H), 7.03 (t, J = 8.7 Hz, 2H), 6.86 (d, J = 8.7 Hz, 2H), 3.81 (s, 3H), 1.33 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 161.7 (d, J = 247.0 Hz), 158.4, 150.2, 136.5, 134.2, 132.0 (d, J = 8.0 Hz), 131.1, 130.1 (d, J = 3.5 Hz), 129.4, 125.3, 124.8, 123.0, 122.0, 119.6, 115.3, 114.8 (d, J = 21.5 Hz), 113.7, 83.4, 55.1, 27.6; ^{19}F NMR (CDCl_3): δ -114.9 (m); HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{26}\text{H}_{24}\text{FNO}_3$: 440.1632, found: 440.1633.

1-tert-Butyloxycarbonyl-2-(4-tert-butyl dimethylsilyloxyphenyl)-3-(4-methoxyphenyl)indole (6d). 4.5 h; yield 91% (0.242 g); hexane $\rightarrow$ hexane : EtOAc (9 : 1, R_f 0.7); ^1H NMR (300 MHz, CDCl_3): δ 8.30 (d, J = 8.1 Hz, 1H), 7.58 (d, J = 7.2 Hz, 1H), 7.37 (t, J = 7.7 Hz, 1H), 7.29–7.23 (m, 1H), 7.14–7.11 (m, 4H), 6.83–6.78 (m, 4H), 3.80 (s, 3H), 1.29 (s, 9H), 1.00 (s, 9H), 0.21 (s, 6H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 158.2, 155.1, 150.4, 136.6, 135.3, 131.5, 131.1, 129.4, 127.2, 125.8, 124.5, 122.9, 121.2, 119.5, 119.4, 115.1, 113.6, 83.1, 55.1, 27.6, 25.7, 18.3, -4.4; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{28}\text{H}_{31}\text{NO}_4\text{Si}$: 496.1915, found: 496.2094.

1-tert-Butyloxycarbonyl-3-(4-methoxyphenyl)-2-(2-thienyl)indole (6e). 10 h; yield 92% (0.186 g); hexane $\rightarrow$ hexane : EtOAc (9 : 1, R_f 0.60); m.p.: 143–145 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.28 (d, J = 8.7 Hz, 1H), 7.55 (d, J = 7.5 Hz, 1H), 7.41–7.28 (m, 5H), 6.99–6.94 (m, 2H), 6.88–6.85 (m, 2H), 3.81 (s, 3H), 1.36 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 158.5, 150.1, 136.7, 134.5, 130.9, 129.14, 129.06, 127.7, 126.5, 126.4, 125.3, 125.1, 124.2, 123.0, 119.8, 115.3, 113.6, 83.4, 55.1, 27.6; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{24}\text{H}_{23}\text{NO}_3\text{S}$: 428.1291, found: 428.1291.

1-tert-Butyloxycarbonyl-3-(4-methoxyphenyl)-2-(2-furyl)indole (6f). 3.5 h; yield 88% (0.172 g); hexane $\rightarrow$ hexane : EtOAc (20 : 1, R_f 0.4); m.p.: 122–123 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.30 (d, J = 8.3 Hz, 1H), 7.59 (d, J = 7.8 Hz, 1H), 7.53 (d, J = 1.7 Hz, 1H), 7.41 (dt, J = 7.9, 1.4 Hz, 1H), 7.30–7.21 (m, 3H), 6.94–6.88 (m, 2H), 6.38 (dd, J = 3.3, 1.8 Hz, 1H), 6.20 (d, J = 3.3 Hz, 1H), 3.83 (s, 3H), 1.45 (s, 9H); ^{13}C NMR (75.5 MHz,

CDCl_3): δ 158.8, 150.0, 145.9, 142.2, 136.7, 130.8, 128.8, 125.5, 125.28, 125.25, 124.9, 123.0, 120.1, 115.4, 113.7, 111.01, 110.96, 83.3, 55.2, 27.8; HRMS (ESI) $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{24}\text{H}_{23}\text{NO}_4$: 412.1519, found: 412.1520.

1-tert-Butyloxycarbonyl-2-(4-carbonitrilphenyl)-5-fluoro-3-(2-thienyl)indole (6g). 11 h; yield 83% (0.174 g); hexane $\rightarrow$ hexane : EtOAc (20 : 1) $\rightarrow$ hexane : EtOAc (9 : 1, R_f 0.45); m.p. 156–158 °C; ^1H NMR (300 MHz, CDCl_3): δ 8.26 (dd, J = 4.8, 9.3 Hz, 1H), 7.69–7.65 (m, 2H), 7.48–7.41 (m, 3H), 7.27–7.26 (m, 1H), 7.15 (dt, J = 8.9, 2.7 Hz, 1H), 7.00 (dd, J = 3.5, 5.1 Hz, 1H), 6.88 (dd, J = 3.5, 1.2 Hz, 1H), 1.29 (s, 9H); ^{13}C NMR (75.5 MHz, CDCl_3): δ = 159.8 (d, J = 241.0 Hz), 149.4, 138.5, 135.3, 132.9, 132.7 (d, J = 1.2 Hz), 131.6, 131.2, 129.7 (d, J = 9.7 Hz), 127.4, 127.1, 125.9, 118.6, 116.7 (d, J = 8.7 Hz), 116.4 (d, J = 4.1 Hz), 113.5 (d, J = 25.2 Hz), 111.9, 105.6 (d, J = 24.8 Hz), 84.6, 27.5; ^{19}F NMR (CDCl_3): δ = -119.6 (m); HRMS (ESI) $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{24}\text{H}_{19}\text{FN}_2\text{O}_2\text{S}$: 417.1079, found: 417.1079.

5-Methoxy-3-(3,4,5-trimethoxyphenyl)-2-(2-furyl)indole (7). The coupling reaction of **5k** (0.5 mmol) with (2-furyl)zinc chloride-LiCl (1.5 mmol) was performed following the general procedure. After 15 h the solvents were removed under reduce pressure, and AlCl_3 (1.5 mmol, 0.5 M solution in THF) were added. After 12 h at ambient temperature the crude reaction mixture was diluted with DCM (10 mL) and a 1 M solution of Rochelle salt was added. Then the organic layer was washed with water (10 mL), brine (10 mL), and dried over MgSO_4 . Column chromatography hexane : EtOAc (9 : 1 $\rightarrow$ 3 : 1, R_f 0.55) gave 48% (0.091 g) of the product; ^1H NMR (300 MHz, CDCl_3): δ 8.59 (br s, 1H), 7.42 (s, 1H), 7.31–7.29 (m, 1H), 7.03 (m, 1H), 6.92–6.89 (m, 1H), 6.78 (s, 2H), 6.40 (s, 2H), 3.97 (s, 3H), 3.86 (s, 6H), 3.82 (s, 3H); ^{13}C NMR (75.5 MHz, CDCl_3): δ 154.8, 153.4, 147.0, 141.2, 137.0, 130.5, 130.2, 129.0, 125.9, 114.2, 113.3, 111.8, 111.7, 107.1, 106.8, 101.0, 61.0, 56.1, 55.8. HRMS (ESI) $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{22}\text{H}_{21}\text{NO}_5$: 380.1493, found: 380.1498.

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